

# Klotho Gene

## Effects on Aging

Allison Langhettee  
BIOL 4097

# Overview

01

## Intro & Background

- What is Klotho gene?
- Discovery/History

02

## Current Status

- Klotho gene structure & function

03

## Effects on Aging

04

## Current Research

05

## Conclusion

06

## Sources



01

# Introduction and Background

# What is Klotho Gene?

- Klotho (KL) gene makes Klotho protein which is very important in phosphate, vitamin D, and calcium regulation within the body
- Klotho is found mostly in kidney cells since the kidneys are responsible for modulating these minerals
- Klotho gene is located on chromosome 13, and the gene itself is around 5.2 kilobase pairs

# Discovery of Klotho Gene

- Klotho Gene was discovered by Dr. Makoto Kuro-o in 1997
- The discovery of the Klotho gene was made accidentally after noticing decreased lifespan and organ failure after silencing this gene in mice models
- They determined that this gene when present would inhibit aging, and when absent would accelerate aging



# Discovery of Klotho Gene

- Kuro-o was interested in studying hypertension, and created a study using transgenic mice to study his hypothesis that increasing  $\text{Na}^+$ -proton transport causes hypertension
- The transgene accidentally disrupted an existing gene, designating this mouse as useless for this study
- However, Kuro-o and his team observed that the mouse began demonstrating signs of accelerated aging
- They determined that the Klotho gene had been disrupted, and the deficiency in Klotho is what had caused the symptoms of aging in the mice

# Discovery of Klotho Gene

- After discovering the Klotho gene from his hypertension study, Kuro-o performed experiments in which he overexpressed Klotho protein
- He found that these mice had a lifespan around 20-30% longer than wild-type mice
- These findings is why Klotho has been coined the “anti-aging gene”

# History of Klotho Gene

- Klotho Gene was named after the goddess Clotho in Greek mythology
- Clotho was the controller of life and was said to decide who lives or dies
- For this reason, when the Klotho gene was discovered to be a determinant in ageing and lifespan, it was given its name in reference to the Greek goddess Clotho





02

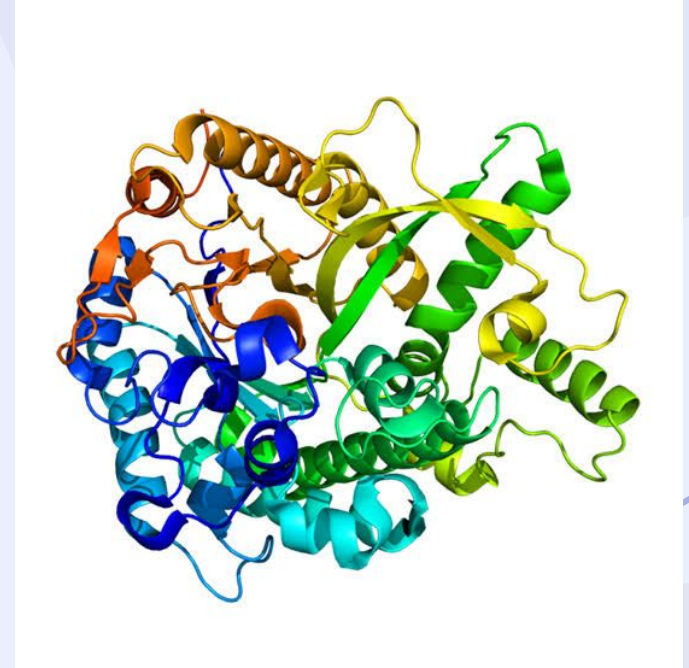
Current  
Status

# Klotho Structure

- Klotho gene makes Klotho proteins, including  $\alpha$ -Klotho,  $\beta$ -Klotho, and  $\gamma$ -Klotho
- Klotho proteins are involved in the fibroblast growth factor (FGF) pathway, which is an important pathway for “embryonic development, tissue homeostasis, angiogenesis, tissue repair,” and other biological process related to growth
- $\alpha$ -Klotho is involved in FGF23 signaling,  $\beta$ -Klotho is involved with FGF19 and FGF21 signaling, and  $\gamma$ -Klotho interacts with FGFR receptors, including FGFR 1b, 2c, and 4
- Our main focus will be on  $\alpha$ -Klotho

# Klotho Structure

- Membrane-bound and soluble forms of  $\alpha$ -Klotho exist
- The membrane-bound form primarily functions as a FGF23 binding site; this regulates phosphate and vitamin D levels
- The soluble form is involved in calcium and potassium regulation through the TRPV5, TRPV6, and ROMK1 channels



# Klotho Gene Function

- $\alpha$ -Klotho functions in several different ways:
  - Role in insulin/IGF-1 pathways
  - Role in Wnt signaling
  - Regulates  $\text{Ca}^{2+}$  homeostasis
  - Regulates Phosphate homeostasis
  - Counteracts oxidative stress

# Klotho Gene Function

- Role in insulin/IGF-1 pathways
  - $\alpha$ -Klotho plays a role in the IGF-1 pathway by decreasing the production of insulin and also increasing the body's sensitivity to insulin
  - Although unclear, some research has also indicated that Klotho is involved with indirectly regulating the insulin receptor substrate (IRS) and IGF-1 receptor
  - It has been proposed that Klotho interacts with forkhead box proteins (FOXOs), increasing their activity and allowing cells to better handle stress and combat aging

# Klotho Gene Function

- Role in Wnt signaling
  - It has been shown that Klotho decreases Wnt signaling, which is a pathway that regulates cell growth and development
  - $\alpha$ -Klotho acts as a Wnt antagonist, which can prevent excessive Wnt signaling and abnormal cell growth
- Regulates Ca<sup>2+</sup> homeostasis
  - Presence of soluble  $\alpha$ -Klotho can increase the presence of the calcium channel TRPV5 on the cellular membrane, which increases the activity of the channel and allowing the kidneys to reabsorb more calcium before it is excreted through the urine

# Klotho Gene Function

- Regulates phosphate homeostasis
  - $\alpha$ -Klotho interacts with the receptor on FGF23, which signals a protein kinase cascade, which ultimately prevents the reabsorption of phosphate in the kidneys, and allows excess phosphate to be excreted in the urine
- Counteracts oxidative stress
  - Very pertinent to our study of aging because oxidative stress can damage DNA and accelerate aging
  - $\alpha$ -Klotho plays a role in the phosphorylation of FOXO3a, which limits generation of reactive oxygen species (ROS) in the mitochondria
  - Increased expression of Klotho also has shown to increase the presence of MnSOD, which is an antioxidant that can neutralize radical byproducts of ROS



03

Effects on  
Aging

# Effects on Aging

- Mutations in the Klotho gene can result in several different conditions consistent with accelerated aging:
  - Chronic kidney disease
  - Hypoactivity
  - Sterility
  - Skin thinning
  - Muscle atrophy
  - Osteoporosis
  - Vascular calcifications
  - Soft-tissue calcifications
  - Defective hearing
  - Thymus atrophy
  - Pulmonary emphysema
  - Ataxia
  - Pituitary gland abnormalities
  - Hypoglycemia
  - Hyperphosphatemia
  - High-plasma calcitriol levels



04

# Current Research

## Current Research

# Gene deletion of Klotho in the dentate gyrus does not affect the number of adult-born granule cells

Patrick Kraus<sup>1</sup>, Monika Marunde<sup>1</sup>, Alexandre Rzyzyski<sup>1</sup>, Emrah Düzel<sup>2,3,4,5</sup>,  
Christoph Kaether<sup>6</sup>, Michael R. Kreutz<sup>1,2,5,7</sup>✉ & Anja M. Oelschlegel<sup>1</sup>✉

# Current Research

- This study by Kraus, et al. looked at how Klotho gene is implicated in adult hippocampal neurogenesis (AHN)
- AHN is the process by which “new functional neurons are generated from adult neural stem cells” in the region of the hippocampus called the dentate gyrus (DG)
- Klotho had been determined to be present in the DG through previous research, but no reports were made as to its necessity in the process of AHN

# Methods

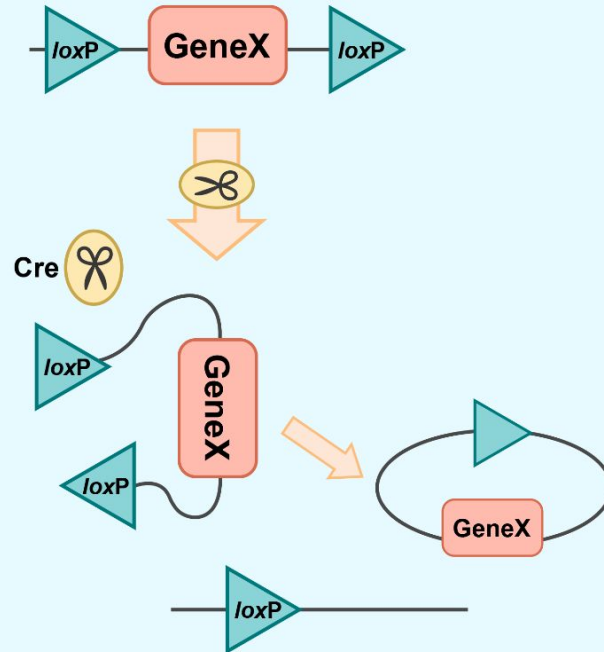
- The researchers created “Klotho DG specific knockout mice” through a POMC-Cre recombinase, which removed Klotho in the DG granule cells
- Fluorescence in situ hybridization (FISH) was used to detect molecules of mRNA; they found a significant decrease in Klotho in DG cells compared to other hippocampal regions
- The researchers then used bromodeoxyuridine (BrdU), a chemical used to help track cell division
- Injections of BrdU were given every 8 hours to track potential neurogenesis
- Mice were then euthanized for brain analysis

# POMC-Cre

- POMC refers to the POMC promoter, which will manage transcription of POMC (pro-opiomelanocortin) gene
- Cre-lox system is a method of altering the genome via the enzyme Cre recombinase
- To create the knockout Klotho gene, the researchers edited the mice genomes to make a Klotho-flox lox/lox
  - Flox indicates the gene being edited (in this case Klotho); A floxed gene is surrounded by lox sites which designate where to cut when Cre is present
- Therefore, our WT mice were POMC-Cre negative (no Cre present so no knockout) and KO mice were POMC-Cre positive (Cre present so knockout gene)

# POMC-Cre

## Deletion



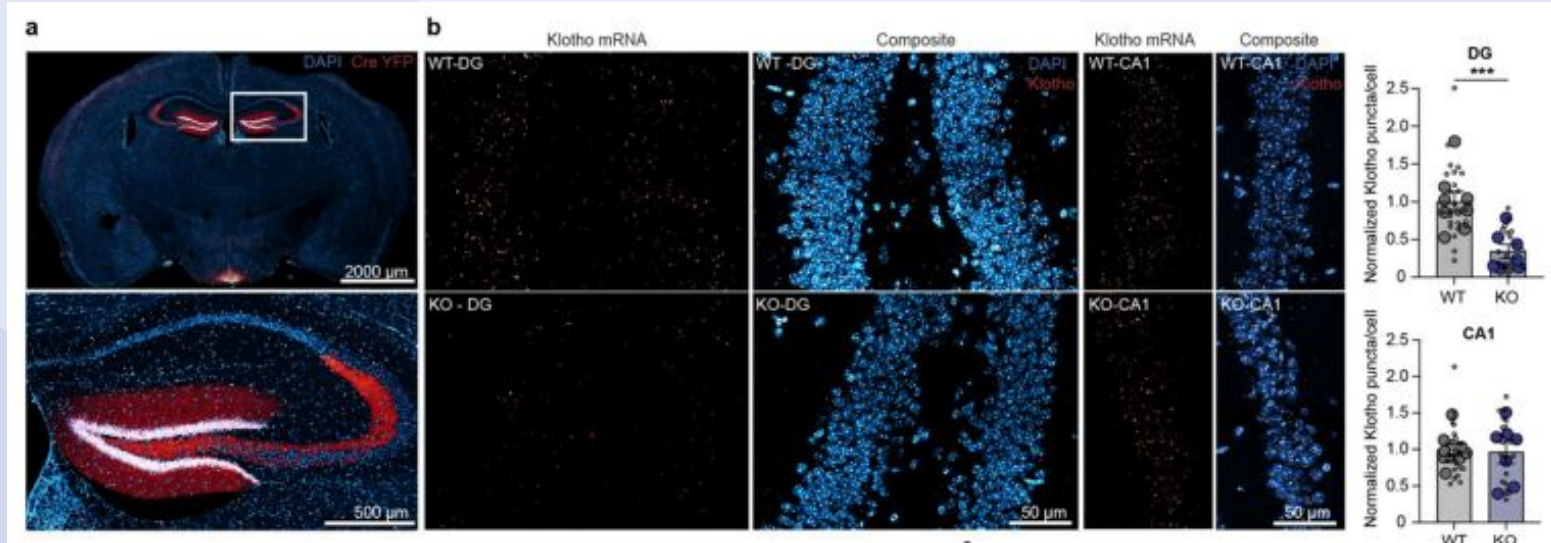
# FISH

- Fluorescence in situ hybridization (FISH) is a method of using dyes to visualize genetic material
- The dye attaches to a specific marker and allows researchers to determine if a certain sequence of DNA is present, or in this case, not present
- In this study, the researchers used FISH to confirm that Klotho mRNA had decreased in the DG as they expected it to, while Klotho mRNA in surrounding hippocampal areas were still present

# Results

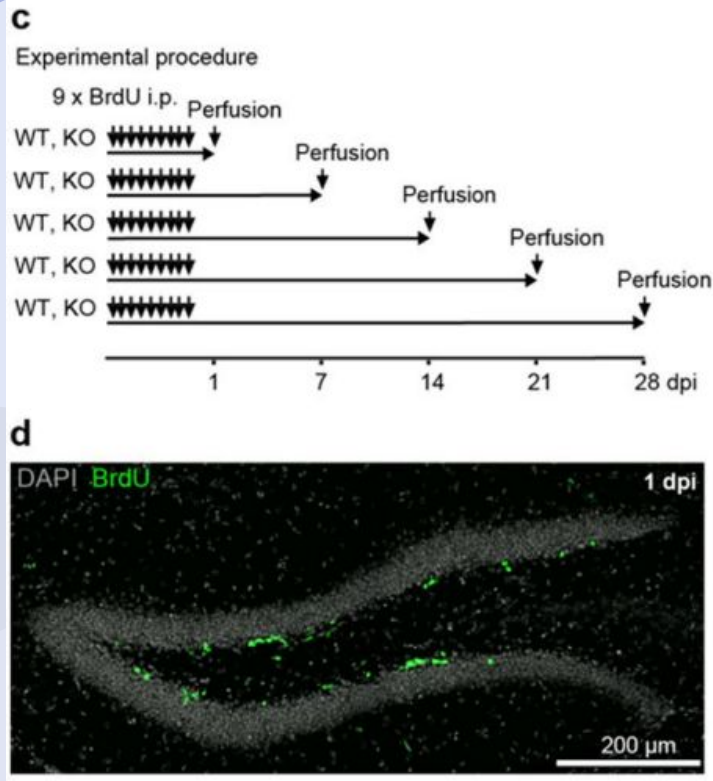
- The researchers found that the Klotho knockout gene in the DG led to observed changes in AHN
- Specifically, the mice with the Klotho-deletion gene had fewer BrdU-positive cells compared to controls for neurons that were 7-10 days old and 14-17 days old
- However, between 21-28 days, there was no observable difference in the number of cells
- The researchers concluded that Klotho in the DG is “required for AHN during a temporally restricted window”, and when removed, more cells are at risk of dying early
- They also found no significant effect of Klotho on stem cells, neuronal progenitor cell proliferation, or gliogenesis

# Results



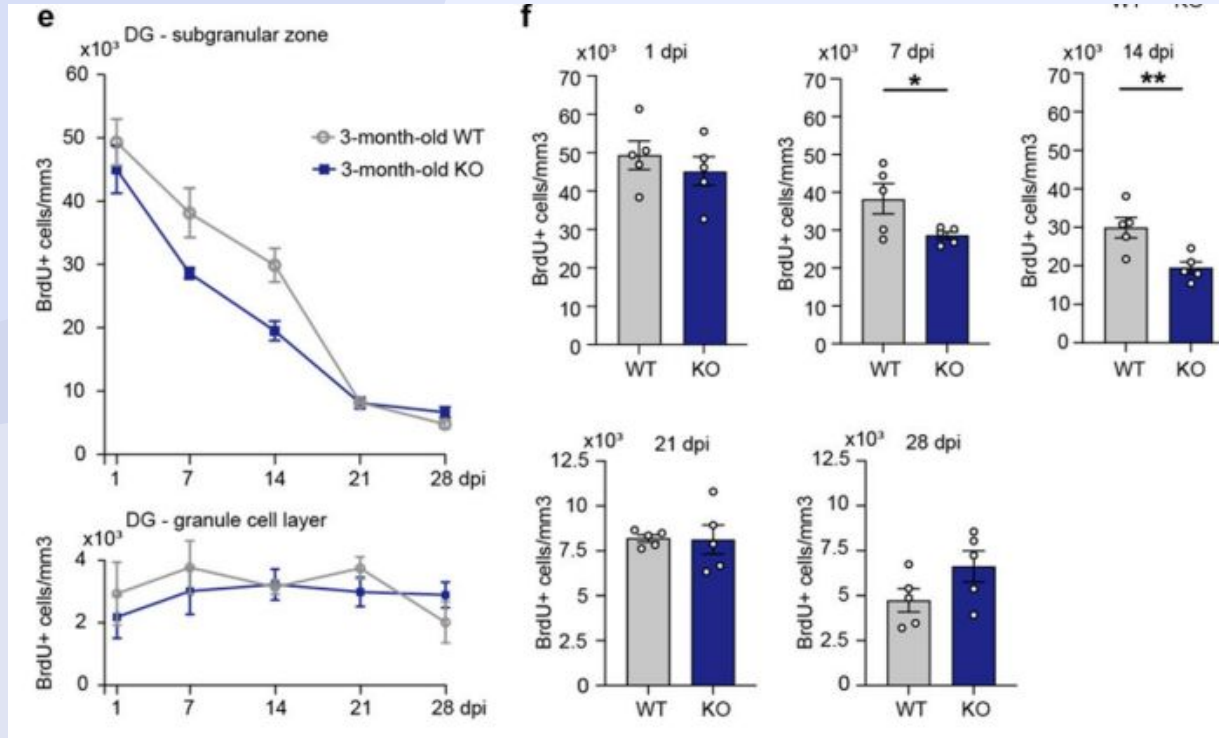
- Fig. 1a shows POMC-Cre in the DG, confirming the knockout gene would be isolated to the DG region
- Fig. 1b shows FISH data that confirms low levels of Klotho mRNA within the DG while no changes in other regions

# Results



- Fig. 1c shows the timeline for BrdU injections and sacrificing the mice to track AHN in several stages
- Fig. 1d shows BrdU-positive levels via immunohistocompatibility (IHC), 1 day post injection

# Results



- Fig. 1e shows survival of neuronal cells over days post injection in subgranular zone and granule cell layer of DG; In KO mice, fewer neurons survived to maturation in early days post injection
- Fig. 1f shows Days 7 and 14 dpi were significant but not significant at 21 or 28 dpi

# Interpretation

- The research by Kraus, et al. tells us in which region and during what time frame the Klotho gene is most important
- Previous studies have referred to Klotho gene as the “anti-aging protein”, but this study shows that Klotho works differently:
  - Aging is tied to a decrease in neural stem cells and progenitor cells, whereas decreased Klotho specifically decreases survival of developing neurons, suggesting a related but not direct tie of Klotho gene to aging
- Because Klotho naturally decreases with age and is associated with cognitive decline, future directions look to how Klotho deficiency could contribute to decreased quality of neurons that survive to maturation



05

Conclusion

# Conclusion

- We started with an introduction of Klotho gene, its discovery, and the story behind its name
- We then discussed the different Klotho protein structures produced by the Klotho gene and the different roles they perform
- We also talked about the different functions of  $\alpha$ -Klotho and its involvement in different pathways, especially its implications for aging
- Next we discussed how mutations in the Klotho gene may lead to symptoms and conditions associating with aging

# Conclusion

- Our discussion continued with current research on the Klotho gene and how it may serve a purpose in adult hippocampal neurogenesis
- From this study we looked at methods, analysis, and results
- Finally, we used the scientific results to interpret Klotho's role in aging and predict where future research may be focused



06

Sources

# Sources

<https://www.ahajournals.org/doi/full/10.1161/01.RES.76.1.148>

<https://en.vectorbuilder.com/resources/vector-academy/lessons/cre-lox.html>

<https://lifesciences.danaher.com/us/en/library/fgf-pathway.html>

<https://medlineplus.gov/genetics/gene/kl/>

<https://mythus.fandom.com/wiki/Clotho>

<https://phys.org/news/2018-03-anti-aging-protein-alpha-klotho-molecular.html>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2637560/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2743784/#S13>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4399270/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4531025/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6611749/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7496967/>

# Sources

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8372657/>

<https://pubmed.ncbi.nlm.nih.gov/25977081/>

<https://www.genome.gov/genetics-glossary/Fluorescence-In-Situ-Hybridization-FISH>

<https://www.jax.org/news-and-insights/jax-blog/2011/september/cre-lox-breeding>

<https://www.nature.com/articles/s41598-026-54703-w>

<https://www.openaccessgovernment.org/ageing-chronic-kidney-disease-biomedical-science/61319/>

<https://www.sciencedirect.com/science/article/pii/S0085253815301277>

<https://www.sciencedirect.com/science/article/pii/S0085253815524605>

<https://www.sciencedirect.com/topics/medicine-and-dentistry/klotho-protein>



Questions/  
Comments?